3. Priority Scheduling

C Program ：

```
#include <stdio.h>

int main() {
    int n, i, j, temp;
    int bt[20], pr[20], wt[20], tat[20];
    float avg_wt = 0, avg_tat = 0;
    printf("Enter Number of Processes: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        printf("\nEnter Burst Time for P%d: ", i + 1);
        scanf("%d", &bt[i]);
        printf("Enter Priority for P%d: ", i + 1);
        scanf("%d", &pr[i]);
    }
    /* Sort according to priority */
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++)   {
            if (pr[i] > pr[j])
            {
                temp = pr[i];
                pr[i] = pr[j];
                pr[j] = temp;
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }
```

```c
    wt[0] = 0;
    for (i = 1; i < n; i++)  {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    printf("\nProcess\tPriority\tBT\tWT\tTAT\n");
    for (i = 0; i < n; i++)  {
        tat[i] = wt[i] + bt[i];
        avg_wt += wt[i];
        avg_tat += tat[i];
        printf("P%d\t%d\t\t%d\t%d\t%d\n",
               i + 1, pr[i], bt[i], wt[i], tat[i]);
    }
    avg_wt /= n;
    avg_tat /= n;
    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
    return 0;
}
```

Shell Program (priority.sh) :

```bash
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i = 0; i < n; i++))
do
    echo "Enter Burst Time for P$((i + 1))"
    read bt[$i]

    echo "Enter Priority for P$((i + 1))"
    read pr[$i]
```

```
done
# Sort according to Priority
for ((i = 0; i < n - 1; i++))
do
    for ((j = i + 1; j < n; j++))
    do
        if [ ${pr[i]} -gt ${pr[j]} ]
        then
            temp=${pr[i]}
            pr[$i]=${pr[j]}
            pr[$j]=$temp
            temp=${bt[i]}
            bt[$i]=${bt[j]}
            bt[$j]=$temp
        fi
    done
done
wt[0]=0
echo
echo -e "Process\tPriority\tBT\tWT\tTAT"
total_wt=0
total_tat=0
for ((i = 0; i < n; i++))
do
    if [ $i -ne 0 ]
    then
        wt[$i]=$((wt[i - 1] + bt[i - 1]))
    fi
    tat[$i]=$((wt[i] + bt[i]))
```

```
    total_wt=$((total_wt + wt[i]))
    total_tat=$((total_tat + tat[i]))
    echo -e "P$((i + 1))\t${pr[i]}\t\t${bt[i]}\t${wt[i]}\t${tat[i]}"
done
avg_wt=$(awk "BEGIN {printf \"%.2f\", $total_wt / $n}")
avg_tat=$(awk "BEGIN {printf \"%.2f\", $total_tat / $n}")
echo
echo "Average Waiting Time = $avg_wt"
echo "Average Turnaround Time = $avg_tat"
```

Output :

```
┌──(ganesh㉿LAPTOP-R3RDH9NG)-[~]
└─$ nano priority.sh

┌──(ganesh㉿LAPTOP-R3RDH9NG)-[~]
└─$ chmod +x priority.sh

┌──(ganesh㉿LAPTOP-R3RDH9NG)-[~]
└─$ bash priority.sh
Enter Number of Processes
3
Enter Burst Time for P1
12
Enter Priority for P1
13
Enter Burst Time for P2
14
Enter Priority for P2
15
Enter Burst Time for P3
16
Enter Priority for P3
17

Process Priority        BT      WT      TAT
P1      13              12      0       12
P2      15              14      12      26
P3      17              16      26      42

Average Waiting Time = 12.67
Average Turnaround Time = 26.67
```